Positional Mechanisms in Attention:
two slots, one scan, and what each one is for

Abstract

Attention encodes no order by itself, and the mechanisms proposed to supply it now number in the dozens. This review argues the space is far smaller than the list.

If the positional map is linear in the query and key, translation-invariant and continuous, the interval operators form a one-parameter matrix group $A(d) = \exp(dM)$, so classifying positional encodings reduces to classifying the generator M up to conjugacy — which leaves exactly two live slots, a real part that scales and an imaginary part that rotates, because $\mathbb{C}$ has exactly two. Defective generators add polynomial factors and nothing else. Writing both slots as one complex log-polar accumulator makes every mechanism a choice of where content enters: magnitude or phase, instantaneous or accumulated, at what sign and what rank. Mechanisms that appear to escape — PaTH, DeltaNet, RWKV-7, CoPE, DAPE, MesaNet, Titans, TAPE, TEM-t, and the classical relative methods of Shaw, Transformer-XL and T5 — escape by breaking one of the three assumptions, and it is possible to say which.

The motivating question is not encoding for its own sake but relational representation: whether learning a structural "where" separately from a sensory "what" is what buys transfer, as TEM argues and MapFormer inherits. Placed in the frame, MapFormer is that factorisation in attention with a parallel scan, and MapEM's Hadamard product is literally TEM's $g \otimes x$ on the tensor product of the two spaces. Measured on environments redrawn at evaluation, a structural code reaches 0.99 where a parameter-matched index code sits on the 0.506 floor.

Two consequences do not follow from any single mechanism. The two slots answer *different questions*: a signed accumulated phase measures net displacement and is a map, a monotone one measures elapsed path and is a clock, and the two are mutually exclusive — so the monotone increments of CARoPE, GRAPE-AP and CoPE are correct on language and disqualifying on navigation. That claim is tested here by crossover: on a task built to want a clock the ordering inverts, and the same unconstrained architecture learns a diffusive accumulator on one task and a ballistic one on the other, which makes the growth exponent a readout of what a task demanded rather than a property of a model. That the distinction is invisible on text is not incidental: in one content-dependent mechanism's own ablation, NoPE attains the lowest perplexity of every encoding tested. And the frame has a second reading, in which the positional operator is an associative memory's transport kernel; under it the fast-weight literature and the positional-encoding literature are seen to have divided the two slots between them without either saying so.

The review closes with measurements on navigation and algorithmic tasks isolating the two axes the frame exposes and nobody varies deliberately — the *sign* of the increment, which costs nothing and is worth more than any other ingredient here, and the *input-side rank* of the content-to-phase map — and with what the two imply for choosing a mechanism for a given task.

1 Introduction

Attention computes content-dependent interactions between tokens and, by itself, encodes no order at all: permute the inputs and the outputs permute with them. A positional mechanism is whatever is added to restore order, and the space of such mechanisms is now large — absolute embeddings, relative biases, rotary phases, decays, gates, content-dependent variants of each, and their state-space analogues.

This review argues that the space is much smaller than the list suggests.

The thesis. If the positional map is linear in the query and key, translation-invariant, and continuous, then the family of interval operators is a one-parameter matrix group $A(d) = \exp(dM)$, and classifying positional encodings reduces to classifying the generator M up to conjugacy. Diagonalising M leaves exactly two live slots — a real part that scales and an imaginary part that rotates — because $\mathbb{C}$ has exactly two. Writing both as one complex log-polar accumulator turns every mechanism in the arena into a choice of *where content enters*: in the magnitude or the phase, instantaneously or accumulated, with what sign and at what rank. Mechanisms that appear to escape this picture escape by breaking one of the three assumptions, and it is usually possible to say which.

What the frame buys. Two things a list of mechanisms does not. First, the two slots turn out to answer *different questions* rather than to be two settings of one dial: a signed accumulated phase measures net displacement and is a map, while a monotone one measures elapsed path and is a clock, and the two are mutually exclusive by construction (§8). Second, the frame makes visible two axes on which the mechanisms genuinely differ and which nobody varies deliberately — the *sign* of the increment and the *input-side rank* of the content-to-phase map — and both are invisible on the tasks the literature evaluates.

Scope, and what this review is not. It covers positional mechanisms as operators on the attention logit: how position enters, what algebra it lives in, and what that implies for what can be computed. It does not cover the frequency-allocation and context-extension line (Position Interpolation, NTK-aware scaling, YaRN, LongRoPE), which varies the readout rather than the operator and is surveyed elsewhere; §12 states only what that line contributes to the frame. Original measurements are reported in §13 and are marked as such throughout.

Positioning. Four surveys of this material were checked. Three — Li (arXiv:2608.10021, August 2026) on absolute and relative methods, RoPE and long-context scaling; the length-extrapolation survey (arXiv:2312.17044); and the infinite-extrapolation framework of Vetcha (arXiv:2601.06113) — contain no content-dependent *phase* mechanism at all: between them, zero occurrences of MapFormer, Selective RoPE, CARoPE, FoX, GRAPE or Mamba-3, and CoPE and Mamba twice each. The fourth, Zhu et al.'s long-context survey (arXiv:2503.17407, March 2025), *does* carry a category for it, "content-aware position embedding", with two members: CoPE and DAPE. That category is the closest existing treatment of this material, and it predates almost all of it — CARoPE by four months, Selective RoPE, MapFormer and GRAPE by eight to nine, Mamba-3 by a year.

So the gap is not that nobody noticed content-dependence. It is that the cell was opened on the *additive* side, by DAPE (2024) and CoPE (2024), and everything that has since filled the *multiplicative* side arrived after the last survey that had a place to put it. The memory reading of §11 and the navigation regime appear in none of the five.

The question this is in service of. The interest here is not positional encoding for its own sake. It is positional encoding as the mechanism by which a model learns a *relational* representation — a "where" — and the claim under test is the one TEM makes and MapFormer inherits: that learning the where, and keeping it *separated* from the "what", is what allows transfer to new environments and to new problems sharing the same relational structure. A sensory code is environment-specific and must be rebuilt every time; a structural code is not, and can be carried over. On that reading the axes below are not descriptive categories but the design space of the where, and §10.1 states what the separation is measured to buy.

What is claimed, and what is not. Three claims, in descending order of how much of the evidence is ours.

C1 (analytical). The arena is a classification rather than a list: under linearity, translation invariance and continuity the interval operators form a one-parameter group, leaving two live slots, and every mechanism that appears to escape breaks one of the three assumptions in a way that can be named. *The classification itself is not ours* — Puranik states it, GRAPE and Mamba-3 independently state the two-slot decomposition. The contribution is the audit: the placement of every mechanism in §9, the escape routes in §9.1, and the observation that the fast-weight literature has been occupying the same two slots from the other side (§11).

C2 (conceptual, and tested). The two slots answer *different questions*. A signed accumulated phase measures net displacement and is a map; a monotone one measures elapsed path and is a clock; they are mutually exclusive by construction. The consequence is that a monotone increment is not a defect — it is correct on language and disqualifying on navigation. This is falsifiable and is tested by crossover in §13.8: on a task built to want a clock the cost of forcing monotonicity falls from 0.28 to nothing measurable, and an architecture free to choose adopts a different accumulator per task. Had the cost been the same on both, the claim would be withdrawn.

C3 (empirical). Two properties of the content-to-phase map that no paper in this literature varies deliberately — the *sign* of the increment and its *input-side rank* — are worth more per parameter than anything else measured here, and both dominate the choice of positional *encoding*.

Not claimed: the taxonomy and the closure argument (C1's caveat above); the content-dependent rotation, which MapFormer, Selective RoPE and Mamba-3 arrived at independently; that a signed increment is generally better, which C2 denies; and that a fixed index code cannot perform contextual counting, which is CoPE's result and is reproduced here in another architecture as an anchor (§13.8).

Organisation. §2–3 set up the polar frame and the classification. §4 places the classical relative methods, which fail the classification's third condition. §5–6 treat the two slots in turn and §7 unifies them in one equation. §8 is the conceptual consequence. §9 is the relational map over eight axes; §10 places two mechanisms from the cognitive-map literature and §11 re-reads the whole frame from the fast-weight side. §12 covers what the long-context line contributes, §13 reports the measurements, and §14 draws out what they imply for choosing a mechanism.

2 The shared frame

Rotary attention transforms the query and key of head h by a block-diagonal rotation before the inner product. Splitting a head's d_{head} coordinates into $n_b = d_{\text{head}}/2$ planes and identifying each plane with $\mathbb{C}$, write $\tilde{q}_t^{(c)}, \tilde{k}_s^{(c)} \in \mathbb{C}$ for the query at position t and key at position s in plane c. The pre-softmax logit is

$$a_{ts} \;=\; \frac{1}{\sqrt{d_{\text{head}}}} \sum_{c=1}^{n_b} \mu_{t,c}^{q}\, \mu_{s,c}^{k}\, \cos\!\big(\theta_s^{(c)} - \theta_t^{(c)} + \psi_{s,c}^{k} - \psi_{t,c}^{q}\big), \tag{1}$$

in polar coordinates $\tilde{q} = \mu^q e^{i(\psi^q + \theta_t)}$, $\tilde{k} = \mu^k e^{i(\psi^k + \theta_s)}$. Every logit is a sum of products $\mu \cos \varphi$.

A positional mechanism is a choice about one of those two factors. RoPE, MapFormer, Selective RoPE, CARoPE and Mamba-3's rotation set the phase φ. PoPE, ALiBi, the Forgetting Transformer and every gated linear RNN set the magnitude μ. xPos, MapPoPE and Mamba-3 set both. That is the whole design space of (1), and §3 explains why it is not an artefact of the parameterisation.

3 The classification, and why there are two slots

Let $\tilde{q}_t = F(t)q_t$ and $\tilde{k}_s = G(s)k_s$, so the logit is $q_t^\top F(t)^\top G(s)k_s$, and impose three conditions:

- **Linearity.** F and G act linearly on q and k.
- **Translation invariance.** $F(t)^\top G(s)$ depends only on $s - t$.
- **Continuity** in the interval.

Normalise so $F(t)^\top G(t) = I$, absorbing the constant into the keys. Then $A(d) := F(0)^\top G(d)$ satisfies $A(0) = I$ and $A(d)A(e) = A(d + e)$: the interval operators form a *one-parameter matrix group*, so $A(d) = \exp(dM)$ for a fixed generator M, and the classification of positional encodings is the classification of M up to conjugacy. This argument is due to Puranik (2026); GRAPE (Zhang et al., ICLR 2026) develops the same framework independently and uses it to prove exact special cases. The two-slot decomposition itself has now been stated at least three times independently: as multiplicative rotations plus additive logit biases (GRAPE, December 2025), as "a multiplicative transformation and an additive bias" indexed by the relative position (Vetcha, January 2026), and as the real and imaginary parts of one generator (Puranik, April 2026). It should be treated as settled rather than as anyone's contribution.

Putting M in Jordan form gives three cases and no others.

- **Diagonalisable, real eigenvalue** λ. The plane contributes $e^{\lambda d}$: exponential decay for $\lambda < 0$, NoPE at $\lambda = 0$, and a blow-up for $\lambda > 0$ that no causal design wants.
- **Diagonalisable, conjugate pair.** The real canonical form is $e^{\lambda d}\,\mathrm{rot}(\omega d)$ — *damped RoPE*. This is the Re / Im split of (1), and **there are exactly two slots because $\mathbb{C}$ has exactly two**. RetNet, xPos and Mamba-3 all live here.
- **Defective.** Repeated eigenvalues with a nilpotent part give polynomial factors d^r times the above. ALiBi is realisable this way after augmenting q and k; Jordan-RoPE (2026) is the deliberate occupant, coupling a complex eigenvalue to a nilpotent in the same block to obtain modes $d^r e^{i\omega d}$.

A remark of Puranik's is worth isolating because it is the bridge to everything content-dependent: a gate with a data-dependent decay should be read as *learning how far to advance time*, not as changing a rate. That is exactly the substitution $t \mapsto S_t = \sum_{u \le t} G(x_u)$ of §7.

4 The classical relative methods

Shaw-style relative position representations, Transformer-XL and T5's relative bias predate the rotary line and are still widely deployed. They are one of the four escape routes of §9, and the one whose route out is least obvious.

Shaw et al. add a learned vector indexed by the *clipped* lag,

$$s_{ij} = \frac{q_i^\top \left(k_j + a^K_{r_{ij}}\right)}{\sqrt{d_{\text{head}}}}, \qquad r_{ij} = \operatorname{clip}(j-i,\,-K,\,K),$$

and T5 compresses the same idea to a per-head scalar, $s_{ij} \mathrel{+}= b_{h,\,\text{bucket}(j-i)}$. Transformer-XL carries the relative form across cached segments.

They are linear in q and k and translation-invariant, and they fail only on continuity — clip and bucket are step functions, so no generator exists and beyond the table's range the operator is constant rather than continuing.

The trade-off is exact and worth stating, because it is the one design choice the rotary line silently makes. **A table can represent an arbitrary function of the lag inside its window and cannot extrapolate outside it at all; a one-parameter group is restricted to the three families of §3 and extrapolates by construction.** RoPE gives up expressiveness over short lags to buy a functional form that is defined at every lag. Whether that purchase is honoured is a separate question, and the long-context literature's own conclusion is that it frequently is not: the ability to *compute* positional features beyond the training length does not imply reliable generalisation there, which is why context extension is evaluated by short-context retention, position-wise perplexity, retrieval and long-context reasoning rather than by whether the encoding is defined.

5 The phase axis: a family of scans

Every content-dependent phase mechanism in the arena maintains a state by cumulative summation and reads it out linearly:

$$s_t \;=\; s_{t-1} + \varphi\big(x_{t-K+1}, \dots, x_t\big) \in \mathbb{R}^m, \qquad \theta_t \;=\; A\, s_t \in \mathbb{R}^{Hn_b}, \tag{2}$$

and they differ in three numbers: the **state rank** m, the form of the increment φ, and its **temporal support** K.

Table 1: The generator family. Every content-dependent phase mechanism maintains a state by cumulative summation and reads it out linearly; they differ in three numbers.

mechanism	m	φ	K	readout A
RoPE	1	$\equiv 1$ (constant)	1	ω
MapFormer	r	$W_{\text{out}}W_{\text{in}}x$	1	$\operatorname{diag}(\omega)$
CARoPE	H	$1/(\operatorname{softplus}(xW)+1) \in (0,1)$	1	geometric
Selective RoPE	Hn_b	$\sigma(W_g x) \odot (\kappa * W_\omega x)$	4	τI

RoPE is the $\varphi \equiv 1$ special case: $\theta_t = t\,\omega$ is a clock that advances by the same amount on every token. Everything else lets the content set the rate.

5.1 Nesting

RoPE $\subset$ MapFormer $\subset$ Selective RoPE, each inclusion *removing a constraint* rather than adding a mechanism.

MapFormer inside Selective RoPE. Four settings suffice: (i) the convolution kernel becomes a delta at the last tap, so K collapses to 1; (ii) the gate bias is set to a constant, making $\sigma(\cdot)$ a scalar g; (iii) the angle projection is set to $A/(\tau g)$ where $A = \operatorname{diag}(\omega)W_{\text{out}}W_{\text{in}}$; (iv) the projection bias is zeroed. The two models then compute the same θ to 1.5×10^{-5} in float32, and the constructed W_ω has rank 2 of 64.

RoPE inside Selective RoPE is unconditional: $W_\omega := 0$ with bias b gives $\theta_t = \tau(t+1)b$, an exact index clock (constant to 7×10^{-9}).

RoPE inside MapFormer costs something, because φ has no bias term: a constant increment requires every embedding row to share its projection onto W_{in}. Taking $W_{\text{in}} = [I_r\ 0]$ and pinning the first r embedding coordinates achieves it exactly, at a price of r of the d embedding dimensions.

Consequently **every comparison inside this column asks whether a constraint is worth keeping, and none of them is about expressivity.**

5.2 The factorisation identity

Cumulative summation is linear, so it commutes with the readout:

$$\theta_t \;=\; \omega \odot \textstyle\sum_{u \le t} W_{\text{out}} W_{\text{in}} x_u \;=\; \operatorname{diag}(\omega)\, W_{\text{out}} \textstyle\sum_{u \le t} W_{\text{in}} x_u. \tag{3}$$

All $Hn_b = 64$ of MapFormer's phase channels are linear readouts of a single r-dimensional accumulator (checked against the implementation at 1.5×10^{-5} in float32). The multi-scale frequency ladder is a *view* of one quantity, not 64 independent clocks — which is what makes the phase a coherent position code rather than a bag of unrelated oscillators.

5.3 Coherence rank

Define the **coherence rank** ρ as the dimension of the affine span of $\{\theta_t\}$, i.e. $\operatorname{rank}[\, A{\cdot}B \mid A\varphi(0)\,]$ when $\varphi(x) = Bx$. The affine part matters: RoPE's constant φ has $B = 0$, so the linear rank alone would read 0. Then $\rho = 1$ for RoPE, r for MapFormer, and generically Hn_b for Selective RoPE.

Two thresholds bracket the useful range. Below, a packing argument: if N^D positions are carried by a ρ-dimensional code, the minimum separation of distinct positions falls at worst like

$$\min_{p \ne p'} \|\theta_p - \theta_{p'}\| \;\lesssim\; N^{-\max(D/\rho,\,1)}, \tag{4}$$

saturating at the lattice's own spacing once $\rho \ge D$. This is an upper bound and is not tight: measured exponents track it at $D \le 3$ and run steeper at $D = 5$ (-2.99 against a predicted -2.50). Above, once $\rho \ge D$ the bound says nothing further, and what remains is a question about conditioning rather than capacity — see §13.

5.4 What $K > 1$ costs

With $K = 1$ and φ linear, (2) is exactly the action of the free *commutative* monoid on tokens: the phase depends on the multiset of tokens traversed and nothing else, so $\theta(a \text{ then } b) = \theta(b \text{ then } a)$ and $\prod_{u \le t} R(\omega\Delta_u) = R(\omega \sum_{u \le t} \Delta_u)$. That collapse is what buys the parallel prefix scan.

A convolution with $K > 1$ makes a token's increment depend on its neighbours, so the phase stops being a function of the multiset and the map from action sequences to $SO(2)^{n_b}$ is no longer a homomorphism. Commutativity is the central computational property here, and it is also the stated limitation: a turn/forward action space, where displacement depends on accumulated heading, is not representable by a fixed per-token increment.

6 The magnitude axis

The other factor of (1). Three occupants, differing in whether content enters instantaneously or by accumulation.

PoPE (Gopalakrishnan et al., 2025) is defined by *deleting* a term. RoPE's real-valued q carries an implicit intrinsic phase ψ^q, and (1) shows it entering the cosine alongside the positional angle — what PoPE calls an entanglement of "what" with "where". PoPE removes the interaction $\psi^k - \psi^q$ and keeps only a content-dependent *magnitude*, $\mu = \text{softplus}(\cdot) \geq 0$, with the phase a pure index clock plus a learned per-(head, frequency) constant δ_c. It is the exact complement of MapFormer, which amplifies the same phase term into a learned accumulator.

Forget gates. FoX computes a scalar $f_t = \sigma(w_f^\top x_t + b_f)$ per head and adds $\sum_{j<\ell\leq t} \log f_\ell$ to the logit. Gated linear RNNs (GLA, Mamba, Mamba-2) do the same thing inside a recurrence, $S_t = \text{Diag}(\alpha_t) S_{t-1} + k_t v_t$. GRAPE proves FoX is an exact instance of its additive family. All are accumulated content-dependent magnitude, with $\log \gamma < 0$ by construction.

ALiBi is the index-driven, content-free member: a linear-in-lag penalty, realisable by a defective generator.

7 One equation

The two axes are disjoint coordinates of a single object. Put each token's contribution in *log*-polar form. Let $c^q, c^k, G : \mathbb{R}^d \to \mathbb{C}^{n_b}$ and set

$$\boxed{\tilde{q}_t = \exp\big(c^q(x_t) - \overline{S_t}\big), \qquad \tilde{k}_s = \exp\big(c^k(x_s) + S_s\big), \qquad S_t = \textstyle\sum_{u\leq t} G(x_u),} \tag{5}$$

componentwise, with $a_{ts} = \frac{1}{\sqrt{d_\text{head}}} \sum_c \text{Re}[\overline{\tilde{q}_t^{(c)}}\, \tilde{k}_s^{(c)}]$. Writing $c = \gamma + i\pi$ and $G = \delta + i\eta$ with all four real,

$$a_{ts} = \frac{1}{\sqrt{d_\text{head}}} \sum_c \underbrace{e^{\gamma^q_{t,c} + \gamma^k_{s,c} + \sum_{t<u\leq s} \delta_{u,c}}}_{\text{magnitude}} \cos\Big(\underbrace{\pi^k_{s,c} - \pi^q_{t,c} + \textstyle\sum_{t<u\leq s} \eta_{u,c}}_{\text{phase}}\Big). \tag{6}$$

c is the instantaneous complex log and G the accumulated one; real parts are magnitude, imaginary parts are phase. Every mechanism in §§5–6 is a choice of which of the four real fields $\gamma, \pi, \delta, \eta$ is non-trivial.

The conjugate is load-bearing, and the two accumulators carry opposite signs. The Hermitian form $\overline{\tilde{q}}\,\tilde{k}$ *subtracts* phases and *adds* log-magnitudes. So $\text{Im}\, G$ must be carried with the *same* sign on q and k to yield $\sum_{t<u\leq s} \eta_u$, while $\text{Re}\, G$ must be carried *oppositely* to yield $\sum_{t<u\leq s} \delta_u$; carried with the same sign it would give $\sum_{u\leq t} + \sum_{u\leq s}$, a joint scale carrying no information about separation. Perturbing the prefix $u \leq t$ and asking whether the score moves — a relative score must not:
Exactly one of the four combinations is interval-relative, and it is the one (5) writes. With $\delta = \log \gamma < 0$ that slot *is* the forget gate; with η content-dependent the phase slot is MapFormer and Selective RoPE. **MapPoPE is not a third mechanism** — PoPE constrains c, the phase family constrains $\text{Im}\, G$, the coordinates are disjoint, so imposing both is an immediate consequence rather than a design.

Table 2: Only one of the four sign combinations is interval-relative. The prefix $u \le t$ is perturbed and the score must not move.

$\mathrm{Re}\, G$	$\mathrm{Im}\, G$	relative prefix leak
opposed	**aligned**	$\mathbf{3.4 \times 10^{-16}}$
opposed	opposed	1.4
aligned	aligned	3.1
aligned	opposed	7.1

Interval-invariance is the right notion of "relative". With content-dependent increments the logit is not a function of $s - t$ and the score matrix is not Toeplitz. The property that survives, and the one that matters, is that the score depends on the tokens in $(t, s]$ and not on the prefix.

8 A map or a clock: what cancellation chooses

It is tempting to read a signed increment as simply better. It is not, and the reason decides which mechanism a task wants.

Attention sees only the interval sum $\theta_s - \theta_t = \omega \sum_{t<u\le s} \Delta_u$. What that quantity *is* depends entirely on whether the increments can cancel:

Table 3: Cancellation chooses what the interval sum measures. The two readings are mutually exclusive by construction.

	Δ signed	Δ monotone
the interval sum is	**net displacement**	**monotone in path length**
two routes $A \to B$	agree	disagree
injective in	**space**, blind to time	**time**, blind to space
what collides	same place, different times	nothing — every instant is distinct
the accumulator is	a **map**	a **clock**

These are mutually exclusive by construction, and each is *correct* for one job. A cognitive map **needs** same-place-different-time to collide — that collision is exactly what revisit prediction asks for. Language needs the opposite: a code that made token 5 and token 15 interchangeable because some "backward" token intervened would be broken, not economical.

So the monotone increment of CARoPE, GRAPE-AP and CoPE is not an oversight that the state-tracking literature caught them in. **On language a monotone clock is the right object**, which is why RoPE — $\Delta \equiv 1$, maximally monotone — is the sensible default there and why nothing went visibly wrong. The limitation appears only where the task demands a quantity that is invariant to route, which on text is essentially parity and formal-language state tracking, and on navigation is everything.

The growth exponent is a readout of which question, not a score. $\alpha \approx 1$ means the accumulator is measuring elapsed time, which is unbounded, so the code must eventually leave any range it was trained on. $\alpha \approx 0.5$ means it is measuring a random-walk position. The ideal for a bounded map would be $\alpha = 0$ — position on a torus is bounded — and nothing here achieves it, because every mechanism accumulates without wrapping. That is a design gap, not a property of the frame.

What MapFormer gave up to get the map. A signed Δ makes two visits to the same cell positionally identical. That is the point, and it also means the phase cannot represent *how long ago*. Anything recency-dependent has to get that information from somewhere else.

> **Which is a candidate account of the forget gate, whose mechanism is otherwise unaccounted for.** (The measurements cited here are set out with their protocol in §13; "unmeasured" there has the specific sense of a contrast inside its minimum detectable effect.) It adds a second, content-dependent, *monotone* accumulator in the magnitude slot — measured at $\alpha = +0.956$, ballistic, beside the phase accumulator's 0.6 — so it restores the clock that the signed phase gave up. Every otherwise-puzzling detail follows: it needs a *live* λ (frozen at zero it lands on the baseline, -0.016, unmeasured) but not *decay*; 5 of 8 seeds learn $\lambda < 0$, and those gain most ($r(\lambda, \text{gain}) = -0.516$). A negative λ still gives a monotone accumulator, counting up instead of down. On this reading it was never a forget gate — it is a clock, and the sign of λ only sets which way it runs.
>
> **Predicted, and untested:** any monotone content-dependent increment of either sign should reproduce the gain; a content-*independent* constant increment (a pure index clock) should reproduce much of it; and the gain should vanish on a task with no recency structure.

This is now tested directly, and it holds. Until recently the dichotomy rested on the construction above plus a correlation across four arms (§13.7); every task measured here wanted a *map*, so "signed wins" and "signed suits this task" were not separable. On a task built to want a *clock*, the ordering inverts as predicted: constraining the increment to be monotone costs 0.215–0.280 on the torus and nothing measurable there, and the same unconstrained architecture learns $\alpha = 0.591$ on one and 0.967 on the other. §13.8.
Read this way the frame's two slots are not two knobs but **two answers**: a signed phase is a map, a monotone accumulated magnitude is a clock, and a system that needs both must spend both slots. Mamba-3 claiming both deliberately is then the natural endpoint rather than a curiosity. In softmax attention the combination is so far only partial: MapPoPE pairs a content-dependent phase with a content-dependent magnitude that is *instantaneous*, so it supplies no clock at all. A content-dependent phase alongside an *accumulated* content-dependent magnitude, in softmax attention, is the configuration nothing in the table occupies.

9 The relational map

Eight axes separate everything in the arena. They are not independent, but no two mechanisms agree on all eight.

- **A1 Factor** — magnitude (Re), phase (Im), both, or neither.
- **A2 Locus** — instantaneous (c) or accumulated (G).
- **A3 Driver** — the index, the token content, a known position, or a query–key *pair*.
- **A4 Algebra** — semisimple diagonal ($\mathbb{C}^\times$), unipotent, non-semisimple, or non-abelian.
- **A5 Sign** — may the increment be negative?
- **A6 Rank** of the content-to-increment map.
- **A7 Host** — softmax attention, or linear attention / SSM.
- **A8 Composition** — parallel prefix sum, path-ordered product, or neither.

Table 4: **The relational map.** Mechanisms that fit (5) or its unipotent extension. **Bold** marks a value held by at most three of the eighteen rows — what distinguishes that mechanism from the field — and nothing else; no row is emphasised as a whole. Note what this does *not* bold: `signed` is held by five rows and is not rare. The claim of §13 is that nobody varies it *deliberately*, which is a different statement. *Jordan-RoPE's magnitude factor is a *polynomial* in the lag, $d^r e^{i\omega d}$: its generator couples a complex eigenvalue to a nilpotent in one defective block. ‡FoX's bias $\sum \log f_\ell$ and GLA's weight $\prod \gamma$ are the same object; the unipotent label follows GRAPE's realisation, which augments q, k to embed the bias in GL, and read as $\mathbb{C}^\times$ it sits with Mamba/GLA. §Scoped: LieRE reads its rotation directly off a known position and its generators need not commute; Algebraic PE is abelian for sequences and grids but not for k-ary trees; GRAPE's additive family is "rank-1 (or low-rank)" in its own abstract, and "full" applies only to its general edge potential. †xPos carries an exponential decay on the magnitude, so its magnitude increment is fixed negative as ALiBi's is; the *phase* sign is omitted throughout for index-driven rows because a global phase sign is a basis convention, whereas a decay and a growth are different objects. ‖**MapPoPE is this work's construction, not a published mechanism**: it is the immediate consequence of imposing PoPE's constraint and MapFormer's together on disjoint coordinates (§7), included because the combination is otherwise unoccupied. Every other row is published. For the Sources list, MapEM is in the MapFormer paper and GRAPE-M and GRAPE-A/AP are the two families of the single GRAPE paper. ¶MapEM evaluates the same equation on the *tensor product* of the content and position spaces rather than on $\mathbb{C}^{n_b}$; see §10.

mechanism	A1 factor	A2 locus	A3 driver	A4 algebra	A5 sign	A6 rank	A7 host
NoPE	—	—	—	—	—	—	softmax
RoPE	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
ALiBi	magnitude	acc.	index	unipotent	fixed < 0	n/a	softmax
xPos	both	acc.	index	$\mathbb{C}^\times$	fixed $< 0^\dagger$	n/a	softmax
PoPE	magnitude	**inst.**	content	—	> 0	full	softmax
MapFormer	phase	acc.	content	$\mathbb{C}^\times$	signed	**rank** r	softmax
MapEM¶	phase	acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
MapPoPE‖	both	inst.+acc.	content	$\mathbb{C}^\times$	signed	rank r	softmax
Selective RoPE	phase	acc.	content	$\mathbb{C}^\times$	signed	full (or 8–16)	both
CARoPE	phase	acc.	content	$\mathbb{C}^\times$	$\in (0,1)$	**1/head**	softmax
GRAPE-M	phase	acc.	index	$\mathbb{C}^\times$	n/a	n/a	softmax
GRAPE-A / AP	magnitude	acc.	content	**unipotent**	≥ 0	1–full§	softmax
FoX	magnitude	acc.	content	unipotent‡	< 0	scalar/head	softmax
Jordan-RoPE	both*	acc.	index	**non-ss.**	n/a	n/a	softmax
LieRE	phase	direct§	**position**	$SO(d)$, non-ab.	n/a	**tile** $b\times b$	softmax
Algebraic PE	phase	acc.	**structure**	$\mathbb{C}^{\times\,\S}$	n/a	n/a	softmax
Mamba / Mamba-2, GLA	magnitude	acc.	content	$\mathbb{C}^\times$	< 0	full/scalar	SSM
Mamba-3	**both**	acc.	content	$\mathbb{C}^\times$	signed	full	SSM

Every *abelian* row above composes by parallel prefix sum — the eighth axis, **A8**, which the table does not have a column for because all but one row share a value. There are two exceptions. LieRE's generators need not commute, so it obeys no prefix-sum law and reads its rotation directly off a position rather than accumulating; and Algebraic PE is abelian for sequences and grids but not for k-ary trees.

9.1 What leaves the frame, and how

Table 5: Mechanisms that leave the frame, and which of the three assumptions each gives up.

mechanism	assumption broken	consequence
PaTH, DeltaNet, RWKV-7	the *fixed basis* (A4 non-abelian)	path-ordered product of $I+$rank-1; no S_t
CoPE, DAPE	*per-token factorisation* (A3 = pair)	$p_{ij} = \sum_{k=j}^{i} \sigma(q_i \cdot k_k)$; no θ_t
MesaNet, Titans, TAPE	the *path structure* (A8)	prefix-global inverse / deep memory / layerwise update
Shaw, T5, Transformer-XL	*continuity* (Puranik's third)	a lookup table over clipped or bucketed lag; no generator M
TEM-t	*linearity* and *commutativity*	$e_t = \sigma(e_{t-1} W_{a_t})$; see §10

Non-abelian transport: PaTH, DeltaNet, RWKV-7. The interval operator becomes a cumulative *product* of Householder-like reflections $I - \beta_t w_t w_t^\top$ rather than a sum of increments. PaTH proves this takes attention beyond TC^0 (on the assumption $TC^0 \neq NC^1$); DeltaNet contributes the chunkwise-parallel algorithm that makes the product trainable at scale; RWKV-7 generalises the rule to vector-valued gating and shows it recognises *all* regular languages with a constant number of layers. The price is that no $A(d)$ and no S_t exist.

Pair-dependence: CoPE. Position stops being a property of a token. A gate is computed for every query–key *pair*,

$$g_{ij} = \sigma(q_i^\top k_j), \qquad p_{ij} = \sum_{k=j}^{i} g_{ik},$$

so the distance from i to j counts only the tokens the query elects to count. The paper's own example states the idea most clearly: to measure position in *sentences*, the gate need only fire on sentence separators. Position becomes query-dependent and denominated in task-relevant units, which is exactly why no per-token θ_t exists and why CoPE cannot be written in (2). It solves selective copying, counting and Flip-Flop, on which standard encodings fail, and improves perplexity on language and code. It is also monotone ($\sigma > 0$), so §8 applies: CoPE builds a better *clock*, one whose tick is chosen by content, and not a map.

DAPE leaves by the same door and one more. It computes an additive bias as $\mathrm{MLP}(a_{ij}, b_{ij})$ from the attention score and a fixed positional prior, so it too is a function of the *pair*, and the MLP costs linearity as well. It is worth naming because of when it appeared: DAPE and CoPE are both 2024, and they made position content-dependent on the **additive** side more than a year before CARoPE, Selective RoPE, MapFormer and Mamba-3 did the same on the multiplicative side.

Broken path structure: MesaNet, Titans, TAPE. These give up increment-accumulation itself. MesaNet solves a local least-squares problem to optimality at every timestep by conjugate gradient — a prefix-global inverse rather than a running sum — and reports lower perplexity than Mamba-2 and other modern RNNs at matched scale while remaining behind transformers on in-context recall, which is the primal/dual trade-off of §11 showing up as a benchmark result. Titans replaces the linear state with a deep memory module trained at test time by gradient descent with momentum and weight decay, and scales past 2M-token needle-in-a-haystack. TAPE keeps a positional *tensor* and updates it across layers through attention and MLP blocks under permutation- and orthogonal-equivariance constraints, improving long-sequence perplexity and length-generalised addition. In none of the three is the operator applied to a stored trace a function of the interval, so the classification does not reach them.

Broken linearity and commutativity: TEM-t. Treated in §10, where the two independent breaks are separated.

9.2 What the mechanisms report

The map says where each mechanism sits; it does not say how well any of them works. Their own headline results, for orientation. These are not commensurable — different scales, data and benchmarks — and are given to indicate what each line takes as evidence, not to rank them.

Table 6: Reported headline results, as stated by each source.

mechanism	as reported
RoPE	the default in most open large language models; the baseline every row below is measured against
CoPE	see §9; its reported results are the toy-task set that motivates it
FoX	outperforms the Transformer on long-context language modelling, length extrapolation and short-context downstream tasks
Selective RoPE	at 370M, **NoPE attains the lowest average perplexity of the five arms** (20.78 against RoPE's 21.42); the content-dependent phase passes RoPE on downstream accuracy only after two further components (47.1 against 46.5, from a 45.8 baseline). Solves parity with a single-layer Transformer
HGRN	its own Table 11: a *data-dependent* phase is worse than a constant one; HGRN2 attributes the complex-recurrence gain to state expansion
Mamba-3	at 1.5B, +0.6 points average downstream accuracy over Gated DeltaNet, +1.8 with the MIMO variant; matches Mamba-2's perplexity at half the state size
LieRE	up to 15.1% relative accuracy over absolute embeddings on video, $\geq 1.5\%$ over RoPE-style, at 0.68% of a ViT-B's parameters
MesaNet, Titans	see §9
MapFormer	1.00/1.00/0.99 on 2D grid navigation (IID / OOD-d / OOD-s) against Mamba's 0.38/0.66/0.30; in 5D MapWM collapses to 0.75/0.50/0.35, *below* CoPE's 0.94/0.80/0.69

Two of these deserve emphasis because they cut against the direction of the field. **NoPE is not a straw man**: in Selective RoPE's own ablation it has the best perplexity of every arm tested, and Kazemnejad et al. showed a causal decoder learns an implicit position without any encoding at all.

And **HGRN's ablation is a direct negative on content-dependent phase** for language, from inside the recurrent literature. Both belong beside the positive results, and both are consistent with §8: on text a clock is all that is wanted, and an index clock is already one.

9.3 Reading the map

(i) The frame's own cells are full. Crossing A1 (magnitude / phase) with A3 (index / content) gives four cells and all four are occupied: RoPE, ALiBi, Mamba-3's rotation, FoX. GRAPE holds the index-multiplicative and content-additive cells; Mamba-3, Selective RoPE and CARoPE hold content-phase. The remaining structure is **content-dependent, signed, orthogonal and still abelian** — contextual multiplicative — which is where MapFormer, Selective RoPE and Mamba-3 sit. CARoPE occupies the same cell on every axis but the sign, where its increment is constrained to $(0, 1)$; §8 is about what that costs.

(ii) A5 is settled next door and unapplied here. Sarrof et al. (2024) observed that SSM gates are "always nonnegative ... due to exponential or sigmoid parameterizations", with a theorem: such models cannot recognise PARITY. Grazzi et al. (ICLR 2025) proved the eigenvalue version and fixed it by extending the range to $[-1, 1]$; PaTH adopts it deliberately; RWKV-7 restricts it only for training stability; and Selective RoPE's §4.2 already demonstrates single-layer Transformer parity from rotations that "model *flips*". Yet CARoPE, GRAPE-AP and CoPE are all monotone — by softplus, by construction, by sigmoid — and none cites any of that literature. §13.1 measures what the constraint costs where it can be seen.

(iii) A6 is thinner than a blank column suggests. Two rows constrain something rank-like for reasons of their own: PaTH's Householder factors are I+rank-1 by definition, and CARoPE emits one scalar per head, putting its content-to-angle map *below* MapFormer's r rather than above it. The nearest deliberate study is LieRE's generator tile-size sweep, which varies density on the *output* side and peaks in the interior at 8×8. What no one sweeps is the *input-side* bottleneck: how many dimensions of content may steer the angle.

(iv) A7 is nearly free. Selective RoPE runs in both hosts and Mamba-3 crosses over by state-space duality, so "softmax versus SSM" separates implementations rather than mechanisms.

10 Where MapEM and TEM-t sit

Two mechanisms from the cognitive-map literature are placeable, and the placements are opposite.

MapEM lifts the frame rather than leaving it. MapEM computes two attention matrices and multiplies them elementwise, $A = \mathrm{softmax}(A_X \odot A_P)$. With $(A_X)_{ts} = q^x_t \cdot k^x_s$ and $(A_P)_{ts} = q^p_t \cdot k^p_s$,

$$(A_X \odot A_P)_{ts} \;=\; \big(q^x_t \otimes q^p_t\big) \cdot \big(k^x_s \otimes k^p_s\big),$$

so the Hadamard product *is* a single bilinear form — on the tensor product of the content and position spaces, of dimension $d_x d_p$ rather than d (verified to 1.9×10^{-6} in float32). Everything in §7 applies with c and G valued in $\mathbb{C}^{d_x d_p}$.

That is sharper than "an AND-gate rather than an OR-gate", and it says what the lift costs: a rank-one tensor $q^x \otimes q^p$ can express content-*and*-position but not content-*or*-position, and there is no fallback channel when either factor is uninformative. It predicts the observed sensitivity to how the position channel is initialised — a single shared p_0 beats separate q^p_0, k^p_0 by $+0.358$ on a matching task (3/3 seeds) — because with no additive fallback a mis-set position factor multiplies the whole score instead of being outvoted.

TEM-t is the same machine with two conditions removed. TEM-t (Whittington et al., 2022) generates position recurrently, $e_t = \sigma(e_{t-1} W_{a_t})$ with σ a ReLU and W_a a general per-action matrix, then sets $Q = K = W_e e$ and $V = W_x x$; content is absent from the logit entirely and enters only through the values.

Its logit is $e_t^\top W_e^\top W_e\, e_s$. Writing $e_s = W_{t\to s} e_t$ and $e_t = W_{0\to t} e_0$, with W_e orthogonal this is $e_0^\top W_{0\to t}^\top W_{t\to s} W_{0\to t} e_0$ — **the interval operator conjugated by the prefix**. Conjugation is trivial exactly when the group is abelian:

Table 7: TEM-t's logit is interval-relative only when the transition group is abelian and there is no nonlinearity — which is MapFormer's configuration.

transition group	logit depends on the interval alone?
abelian (block rotations), no σ	**yes**
general $SO(d)$, no σ	no
abelian, with ReLU	no
general $SO(d)$, with ReLU	no

MapFormer is precisely the surviving row: $SO(2)^{n_b}$ is abelian by construction and there is no nonlinearity on the path. TEM-t breaks both, and either break alone suffices.

This is the mechanical account of a large empirical cost. No group law means no S_t, hence no parallel prefix scan, hence a sequential loop: at $L = 2048$ a faithful TEM is $120\times$ slower than at $L = 128$ and $1632\times$ slower in absolute terms than a MapFormer of $20\times$ *more* parameters. The constant includes interpreter overhead; the shape is the missing group law. TEM-t is 2022 and MapFormer 2025, so on this axis the later contribution is the *restriction* — abelian and linear — that buys the scan and the relative law.

10.1 Factorisation, and what the separation buys

TEM's thesis is a factorisation: structural knowledge g is environment-invariant, sensory input x is environment-specific, and memory stores the conjunction $p = g \otimes x$. Transfer follows because g survives a change of environment while x does not.

The results above say that MapFormer is that factorisation carried into attention with a parallel scan. The accumulated phase $\theta_t = \omega \sum \Delta_u$ is g: identical in any environment with the same transition structure, because it depends on the action stream and not on what was seen.

But that separation is learned, not built. It is worth being exact here, because it is the difference between a factorisation the architecture guarantees and one it merely tends to find. MapFormer's increment is computed from *every* token embedding — $\Delta_u = W_{\text{out}} W_{\text{in}} x_u$ with no gating by token type — so the model has to *discover* that $\Delta \approx 0$ on observations. It does: on a trained model actions move the phase about five times as much as observations do. And the recency result of §13.8 is the same phenomenon isolated — an unconstrained increment learns a content gate in 8/8 seeds, and a magnitude-matched intervention shows the gate is most of the mechanism rather than a component of it. So the separation is real and measurable; it is emergent rather than enforced.

And that is the design, not an oversight. MapFormer's own text is explicit: "to learn a cognitive map, the model has to identify that action tokens update the internal position, while ensuring that observations leave the structure untouched. This is achieved through a two-stage projection process" — the low-rank $W_{\text{out}} W_{\text{in}}$ *is* the identification mechanism, with W_{in} mapping each token to "a low-dimensional action representation $\Delta_t \in \mathbb{R}^r$ . . . capturing the actual motion to perform". Hard-coding the split would remove exactly what the mechanism is for, and would confine it to settings with a labelled action stream — which excludes text, where there are no action tokens and RoPE is the $\Delta \equiv 1$ special case.

This has a consequence for §13.2 worth stating: **the bottleneck is the separator**. The rank r is not merely the dimension of a displacement code, it is the channel through which the what/where distinction has to pass, so the finding that $r = 2$ learns a skewed basis and $r = 4$ repairs it for 384 parameters is a finding about the *separation* and not only about the geometry. An enforced split is therefore an oracle *control* rather than a proposal: it upper-bounds what the learned separation can reach, and the gap measures what learning the split costs. If the two match, the design is vindicated with a number. Content is x. And the conjunction is exactly the WM/EM distinction — additively in the score for MapFormer-WM, and for MapEM *literally* the tensor product, since $(A_X \odot A_P)_{ts} = (q_t^x \otimes q_t^p) \cdot (k_s^x \otimes k_s^p)$. TEM's $g \otimes x$ and MapEM's Hadamard product are the same object; what MapFormer adds is that the interval operator composes by prefix sum, so the where can be built in $O(\log T)$ instead of recurrently.

Why the measurements are transfer measurements. This is easy to miss in the tables above and is the reason they are set up as they are. **Every evaluation here is on an environment the model has never seen**: the observation map is redrawn from a held-out seed, so the entire what-to-where binding is new, and only structure can carry over. Against a measured always-predict-blank floor of 0.506:

Table 8: Transfer to a new environment with the same relational structure. In every row the sensory map is redrawn at evaluation, so nothing environment-specific survives; the gap is what the structural code carries.

task	structural code	index code	floor
torus revisit, fresh map (n=8)	**0.994** / 0.987 / 0.967	0.530 / 0.509	0.506
blind continuation, 64^2 (n=5)	$\mathbf{0.730 \pm 0.247}$	0.154 ± 0.018	0.063
blind continuation, 128^2 (n=3)	$\mathbf{0.823 \pm 0.043}$	0.192 ± 0.022	0.063
compositional motif transfer (n=8)	$\mathbf{0.415 \pm 0.096}$	0.216 ± 0.004	≈ 0.072

An index code sits at or near the floor on a fresh map. It is not that it predicts badly — it cannot locate itself, so it has nothing to bind an observation *to*, and the map it built in the previous environment is worthless. A structural code arrives with the where already valid and needs only to rebind. That is the factorisation claim, measured.

The separation is what is doing the work, not the architecture. Two controls make this attributable. The path-integrated and index arms here are parameter-matched to within 0.4% and differ only in whether the phase is driven by the action stream or the index, so the gap is the *driver*, not capacity. And the blind-continuation result survives context destruction — shuffling the explore-phase observations takes it from 0.918 to 0.074 — so the model is using the conjunction it built in context and not a statistical regularity of the task.

Two boundaries, both of which the factorisation predicts. The separation buys nothing when the structural code cannot be *formed*: under rotation-based actions the displacement depends on accumulated heading, a fixed per-token increment cannot represent it, and the measured effect falls from +0.438 to +0.050 — an index code then does as well, because neither has a usable where.

And it buys nothing when there is no structure to share: the effect is a threshold in map extent (-0.010, $+0.015$, $+0.305$ for 32, 128, 512 occupied cells), so on a small environment, where an episode revisits everything anyway, there is nothing for a structural code to generalise *from*.

What is not shown. These are transfers to a new *instance* of the same structure — a redrawn observation map on the same torus, new motif instances in the same grammar. Transfer across a change of *structure* — a different topology, a different action space, or a graph learned in one environment reused in another — is the claim TEM makes most strongly and is not tested here. So is *faster* learning: everything above is final accuracy, and no sample-efficiency curve has been measured. Both are the obvious next experiments, and §15 lists what the frame predicts for them.

11 The memory reading: what the frame looks like from the other side

A large literature — fast weight programmers, linear attention, gated linear RNNs, state-space models and their hybrids — studies the same equation while asking a different question. It is worth stating the correspondence, because the two literatures partition the design space almost perfectly and neither says so.

One primitive, two forms. The outer product $v_u \otimes k_u$ binds a value to a key, and superposing bindings gives an associative memory read by a query. The object appears as Kohonen's correlation-matrix memory (1972), Schmidhuber's fast weight programmer (1992), the recurrent form of linear attention (Katharopoulos et al., 2020), the hidden state of a scalar-A state-space model, Smolensky's tensor-product role–filler binding (1990), and — via the dual form of Irie et al. (2022) — the weight matrix of any linearly-parameterised layer trained by gradient descent. Schlag et al. (2021) made the linear-attention correspondence explicit. This gives every mechanism two readings:

$$\underbrace{W_t = \textstyle\sum_{u \le t} A(t-u)\, v_u k_u^\top, \quad y_t = W_t q_t}_{\textit{primal}\text{: compress, bounded state}} \qquad \Longleftrightarrow \qquad \underbrace{a_{ts} = \tilde{q}_t \cdot \tilde{k}_s \text{ over kept items}}_{\textit{dual}\text{: keep and attend}}$$

The positional operator is the memory's transport kernel. $A(d)$ is what happens to a stored trace as it ages d steps, and §3 says there are two things it can do:

Table 9: The two slots, read as a memory. $A(d)$ is the transport kernel applied to a stored trace as it ages.

slot	positional reading	memory reading
$\mathrm{Re}\, G$	magnitude / decay	**how fast the trace is forgotten**
$\mathrm{Im}\, G$	phase / rotation	**how the trace is re-addressed as it ages**

It also identifies A8 with the primal/dual choice: **a positional mechanism admits a parallel prefix scan exactly when the memory admits a bounded recurrent state**, because both are the group law $A(d)A(e) = A(d+e)$. That is one condition, not two.

The two literatures have divided the slots without noticing. The fast-weight literature owns $\mathrm{Re}\, G$ exhaustively. Irie and Gershman's review (TMLR 2025) categorises the whole modern family by *decay type* — RetNet a constant scalar, Mamba-2 a scalar, GLA a rate per row of the fast weight — and the words "rotary", "RoPE" and "positional" do not occur in it. The positional-encoding literature owns $\operatorname{Im} G$ and treats decay as a side-condition. Mamba-3 is the first design to claim both deliberately, and it arrives from the memory side, stating the crossing as a proposition on "data-dependent RoPE equivalence". The hybrid quadratic-linear transformers of Irie et al. (2025) use RoPE, but only as an off-the-shelf component inside the attention half rather than as a design axis.

Positional mechanisms are key-side by construction, which makes the frame a theory of addressing. Gershman, Fiete and Irie (2025) argue that what distinguishes key–value memory from classical similarity-based retrieval is that it holds *separate* representations for storage and for retrieval, so the two can be optimised independently — fidelity for values, discriminability for keys — and that the binding constraint on memory performance is addressing rather than capacity.

Every mechanism in Table 4 acts on q and k and none of them touches v. Read through the correspondence above that is not a convention: **a positional mechanism is an addressing rule**, and $A(d)$ says how the address of a trace changes as it ages. The two slots are then two ways for an address to move: $\operatorname{Im} G$ transports it around a torus, so a trace becomes retrievable again at a predictable displacement; $\operatorname{Re} G$ makes it fade, so the trace becomes unretrievable at a rate the content chooses. Nothing in the frame can alter what was stored, only where it can be found.

Why decay is compulsory on one side of the line and optional on the other. A visible asymmetry in Table 4 is that every SSM row carries $\operatorname{Re} G < 0$ while most softmax rows leave the magnitude slot empty. Schlag et al. (2021) supply the reason: an additive outer-product memory has a finite *capacity*, and once the sequence length exceeds it the model enters an overcapacity regime in which stored associations interfere. In the primal form decay is therefore capacity management and not a modelling preference — something has to evict. In the dual form the items are kept separately and the softmax renormalises, so nothing saturates and a decay term (ALiBi, FoX) is an inductive bias one may decline. The same equation, and the slot is mandatory on one side and free on the other.

That also sharpens what the delta rule is for. Schlag et al. introduce it precisely because purely additive updates cannot *edit* a stale association, only bury it; the rewrite is a capacity fix before it is an expressivity fix, and the expressivity — state tracking, parity — comes along with it.

The non-abelian family is the rewritable-memory family. The first row of the out-of-frame table is not a collection of exceptions. DeltaNet, PaTH and RWKV-7 replace the *additive* Hebbian update with a *delta rule*,

$$W_t = W_{t-1} + \sigma(\beta_t)\big(v_t - W_{t-1}\phi(k_t)\big) \otimes \phi(k_t),$$

whose state-transition matrix $I - \sigma(\beta_t)\phi\phi^\top$ is a Householder-like reflection rather than a scalar. Memory becomes rewritable — stale associations are corrected instead of accumulated — and the price is exactly the group law: reflections do not commute, so no $A(d)$ and no S_t exist and the composition is a path-ordered product. **What that family trades away is the positional structure, and what it buys is a memory that can overwrite.** It is also where the sign axis of §9(ii) is deliberately exercised: with $\sigma = 2\,\mathrm{sigmoid}$ the eigenvalues lie in $(-1, 1)$ *including negative values*, which is what unlocks parity and modular state tracking.

And MapEM's lift is a binding. The tensor product $q^x \otimes q^p$ of §10 is Smolensky's role–filler binding: content as filler, position as role. Read from the memory side, MapEM stores each observation bound to its place rather than superposing the two channels additively — which is the same trade the primal/dual axis makes, one level down.

This is a change of reading, not of mechanism. It says where to look for the unoccupied designs: the cells that are empty in one literature are usually the ones the other has already filled.

12 The readout axis

§5 treats the readout $A = \mathrm{diag}(\omega)$ as fixed. A large and practically important literature varies exactly that, and it is disjoint from everything above: Position Interpolation, RoPE scaling laws, NTK-aware scaling, Dynamic NTK, NTK-by-parts, YaRN, LongRoPE and LongRoPE2 all rescale the frequency ladder so a model trained at one length runs at another. Every one of them is index-driven, so on Table 4 they occupy a single cell — but they vary something the eight axes do not name.

The critical dimension. The argument that organises that line is worth importing. Channel r has wavelength $T_r = 2\pi b^{2r/d_{\text{head}}}$. High-frequency channels complete many cycles inside the training length L_0 and are trained through their whole phase range; low-frequency channels may not complete even one. The boundary

$$r_{\text{crit}} \;=\; \min\{\, r : T_r \geq L_0 \,\}$$

separates them, and beyond the training length the channels past it enter phase intervals never seen in training. For LLaMA-2 ($d_{\text{head}} = 128$, $L_0 = 4096$) that is 36 of 128 dimensions. This is a mechanism for length-extrapolation failure that is about *which channels were trained*, not about capacity or precision.

13 Measurements

The axes above are separable only where position must be reconstructed from the input rather than read off the index, which is why the measurements below are on navigation and algorithmic tasks rather than on text.

Setup. The primary task is **torus revisit prediction**: an agent takes actions on an $N \times N$ torus, observations are aliased across cells (16 types over N^2 cells, half of them blank), the token stream interleaves actions and observations, and the model predicts the observation at *revisited* cells only. Unless stated otherwise: $N = 64$, one layer, two heads, $d_{\text{model}} = 128$, $n_b = 32$, trained at sequence length 128 for 300 epochs with warmup and cosine decay at $\mathrm{lr} = 10^{-3}$, and evaluated on a held-out observation map. "Accuracy" is next-token accuracy at revisited positions, against a measured always-predict-blank floor of 0.506. Two further tasks appear where noted: **parity** at sequence length L, and **blind continuation**, in which the model explores with observations revealed and then predicts the observation at each cell with observations withheld (chance $1/16 = 0.0625$).

Standard errors are over seeds; MDE is $2.8\,\mathrm{sd}/\sqrt{n}$, and a contrast inside its MDE is reported as *unmeasured* rather than as a null. Contrasts are paired per seed, and arms compared against one another were trained in one batch.

What these measurements assert. Three findings and one negative, each with the contrast that would have refuted it.

- **F1. The sign of the increment is load-bearing where the task needs a map, and free.** At matched loss and identical parameter count, a signed increment beats an index code by 0.12–0.20 (12/12 seeds); a monotone one beats it nowhere (§13.1). *Refuted by* a monotone arm matching the signed arm on the map task.
- **F2. Rank $r = 2$ fails by conditioning, not capacity.** $r = 4$ buys +0.085 for 384 parameters and is flat to $r = 32$; the learned code at $r = 2$ is skewed rather than small (§13.2). *Refuted by* a capacity slope, or by the deficit growing with world dimension — which was tested, and it does not.
- **F3. One quantity accounts for F1 and F2, and it reads off the task.** The accumulator's growth exponent α correlates with cancellation at $r = +0.9995$ across arms (§13.7), and an unconstrained architecture moves it from 0.591 to 0.967 between a map task and a clock task while every constrained arm stays fixed (§13.8). *Refuted by* the free arm learning the same accumulator on both.
- **N1. The axis on which most mechanisms help is not explained.** Four separate mechanisms help at extrapolated length; α accounts for two and demonstrably not the others, and the imported long-context account is refuted outright (§13.6). This is the gap, stated as one.

The remaining subsections are supporting rather than load-bearing: what the neighbouring constraints are worth (§13.3), why these axes are invisible on text (§13.4), and a cost/benefit summary (§13.5).

13.1 F1: the sign is load-bearing, and free

Six arms × 12 seeds, one batch, $r = 4$ throughout, **parameter count identical at** 204,757 — the constraint adds nothing and does not touch the rank.

Table 10: The sign ablation, **raw** accuracy. Six arms × 12 seeds, one batch, identical parameter count (204,757). The construction control `Vanilla_r4`, omitted here for width, is within +0.006 of `signed` at every length — *unmeasured*, which is the condition for the batch to be readable at all. Read this table beside the loss-matched contrasts below: raw and loss-matched point the same way for the signed-vs-monotone comparison and opposite ways for monotone-vs-index.

arm	increment	T=128	T=512	T=1024	final loss
signed	$W_{\text{out}} W_{\text{in}} x$	1.000	0.978	0.922	0.0002
$\lvert \cdot \rvert$	$\lvert W_{\text{out}} W_{\text{in}} x \rvert$	0.946	0.675	0.558	0.171
softplus	GRAPE-AP form	0.977	0.798	0.584	0.068
CARoPE	$1/(\text{softplus}(\cdot) + 1)$	0.900	0.809	0.645	0.293
RoPE	index	0.799	0.449	0.345	0.784

The raw column is not the contrast of interest, and says so plainly: the monotone arms fit far worse (0.171, 0.068, 0.293 final loss against 0.0002), so a raw comparison against an index arm that fits worse still (0.784) reports convergence rather than representation. **At matched training loss, signed path integration beats an index code by** +0.123 **and** +0.195 **at** $T = 512$ **and** $T = 1024$ **(12/12 seeds), while monotone path integration beats it nowhere** — −0.028 at $T = 128$ and unmeasured beyond. Note also that the index arm sits at or below the measured always-predict-blank floor of 0.506 at both OOD lengths, so its raw margins are not interpretable as skill. The entire measured value of a content-dependent phase over a plain index clock, on this task, requires the increment to be signed.

The learned code shows why. The opposition score $\|\Delta(+x) + \Delta(-x)\|/\overline{\|\Delta\|}$ is 0 when opposite actions cancel exactly and 2 when they are identical:

Table 11: The learned action code. Opposition is 0 when opposite actions cancel exactly and 2 when they are identical.

arm	oppose$_x$	oppose$_y$	$\lvert\cos(N, E)\rvert$
signed	**0.125**	**0.106**	0.218
$\lvert\cdot\rvert$	1.849	1.855	0.587
softplus	1.905	1.885	0.723
CARoPE	**1.981**	**1.977**	**0.930**

CARoPE's published parameterisation reaches 1.98 of a possible 2.00, with its north and east axes all but collapsed onto one another. These models do not merely score worse — they cannot represent a -1 action, and the probe confirms they do not.

At the training length the signed baseline is at 1.000 ± 0.000, so accuracy there cannot show a deficit of any size. The training-length effect is visible in the *loss* instead: every constrained arm fits strictly worse on 12/12 seeds at identical parameters.

This is a replication in a new regime. The constraint and its consequence for parity are established (§9(ii)); what is new is navigation rather than formal languages, and an isolation in which $|\Delta|$ against Δ is a single operation at identical parameter count.

13.2 F2: rank $r = 2$ fails by conditioning, not capacity

The bottleneck r is meant to be the dimensionality of the action space — for a 2D grid, a 2-vector — which makes $r = 2$ sufficient *by construction*. It is not sufficient in practice. Against $r = 2$ at 8× the training length ($T = 1024$ against a training length of 128; 8 seeds):

Table 12: The rank sweep against $r = 2$, 8 seeds. A step at $r = 2$, flat from $r = 4$ to $r = 32$.

	params added	T=512	T=1024
$r = 4$	+384	+0.038	+**0.085** (8/8, t 3.57)
$r = 8$	+1,152	+0.037	+0.091
$r = 16$	+2,688	+0.028	+0.079
$r = 32$	+5,760	+0.038	+0.095

This is a **step at $r = 2$, not a capacity slope**: flat from $r = 4$ to $r = 32$. The cause is conditioning, not capacity. Given four dimensions the model still places its actions in a 2-plane (100.0% of spectral energy; 99.96% even at $r = 32$), vindicating the dimensional argument — what fails is the *basis* that $r = 2$ is forced into:

Table 13: The learned basis at $r = 2$ and $r = 4$. The deficit is conditioning, not capacity.

	opposition (N+$S \approx 0$)	$\lvert\cos(N, E)\rvert$	obs/action norm
$r = 2$	0.495	0.783	0.139
$r = 4$	**0.092**	**0.175**	0.042

At $r = 2$ opposite actions fail to cancel by half the action scale and north and east are nearly parallel. The seed standard deviation also falls from 0.064 to 0.012 at $T = 1024$, which matters because it sets every detectable effect size.

Two further observations. The deficit does *not* grow with the dimensionality of the world: at $D = 5$, $r = 2$ is unimpaired (0.896, its best of three conditions), so the packing account of (4) does not explain it. And an unconstrained generator cannot be projected below rank ≈ 16 post hoc while $r = 4$ trains fine — trained-at-rank and truncated-to-rank are different objects.

13.3 Supporting: what the neighbouring constraints are worth

Selective RoPE's three additions to MapFormer's generator, measured on the torus at $T = 1024$ and on a parity task at $L = 128$; increments against MapFormer at $r = 2$.

Table 14: Selective RoPE's additions to MapFormer's generator, against MapFormer at $r = 2$.

constraint removed	params	parity L=128	torus T=1024
$K = 1 \to 4$ (causal conv)	+193	−0.020	−0.064 (t −1.99)
$m = r \to Hn_b$ (full rank)	+7,873	−0.020	+0.058 (7/8)
φ linear → gated	+8,193	−0.030	+0.086 (8/8)
all three	+16,385	−0.009	+0.048 (t 1.36)
$r = 2 \to 4$, *constraint kept*	**+384**	—	+**0.085** (8/8)

Two readings. First, **the two ≈8k-parameter knobs are statistically indistinguishable from each other** (+0.018 and +0.028, both inside their MDEs), so what they buy is most plausibly capacity rather than either named mechanism — and widening the bottleneck MapFormer already has buys the same thing for 384 parameters, 21× cheaper. Second, the convolution is the only knob negative on both tasks, consistent with §5.4: $K > 1$ costs the homomorphism.

These arms replace the readout $\mathrm{diag}(\omega)W_{\mathrm{out}}$ by τW_2. For the conv and gate rows the function class is preserved exactly — it is the construction of §5.1 — so the difference is initialisation (a geometric frequency ladder against a default draw) plus one scalar; those effects are attributable up to initialisation. The two full-rank rows also add weight normalisation and a bias, so their function class genuinely differs.

A separate probe rules out the natural mechanism for the gate. If the sigmoid worked by suppressing observation tokens — which MapFormer must instead learn to send to $\Delta \approx 0$ — the gate would be near zero on observations. It is 0.416, and the action/observation ratio is 1.35× on the torus where the gate helps and 1.54× on parity where it hurts.

13.4 Supporting: why these axes are invisible on text

Every paper in §9 evaluates on language, recall, state tracking, or — for LieRE and Algebraic PE — images, video and trees. None runs grid navigation, and it is the regime in which these axes are distinguishable at all.

The basic separation is large. On a task where the model explores with observations revealed and then continues blind, predicting the observation at each cell, path integration scores 0.730 ± 0.247 ($n = 5$, 64^2) against 0.154 ± 0.018 for an index code, chance 0.0625, with no seed overlap. The axis is path integration and not the encoding: PoPE with path integration scores 0.847 against 0.117 for PoPE with an index (both $n = 3$; only the MapWM pair was extended to $n = 5$, and it fell 0.158 when it was, so read the ratio and not the values). On the standard revisit task, re-derived from the per-seed data ($n = 8$, held-out map), the position main effect is +0.461 (8/8 seeds, MDE 0.027) and the encoding main effect is +0.003, which is below its own MDE of 0.029 at 5/8 seeds. No ratio is quoted: with a denominator that close to zero the quotient is not a stable quantity. The defensible statement is that one factor moves the result by roughly half and the other does not move it measurably.

Two boundaries are worth stating. The effect is not a property of navigation as such but of *map extent*: at matched observation aliasing it is −0.010, +0.015 and +0.305 for 32, 128 and 512 occupied cells — a threshold, not a gradient ($n = 3$ per condition; the 512 endpoint that makes it a threshold is the least replicated point, though the 0.290 jump is roughly twice the measured noise floor). And it does not survive rotation-based action spaces: under turn/turn/forward, where displacement depends on accumulated heading, the effect falls from +0.438 to +0.050, which is §5.4's commutativity limitation made concrete. Recording the absolute displacement instead of the commanded turn restores it to +0.488.

13.5 Supporting: what each ingredient costs and buys

Table 15: Anchors: what each ingredient costs and buys on the torus task.

mechanism	sets	parameter cost	measured effect
Path integration	phase θ	+448 on 199,042 (+0.23%)	+0.078 parity (16/16); +0.461 torus
Signed increment	sign of Δ	**0**	largest per parameter: removing it costs −0.215/−0.280 at T=512/1024 (12/12)
Bottleneck rank	state rank ρ	+384 (+0.19%)	+0.085 at T=1024 for r=2 → 4 (8/8); flat to r=32
PoPE	magnitude μ	+320	+0.027 *with* path integration; both arms on the floor without it

13.6 N1: the length axis, and a refuted import

Transposed to a content-dependent phase the coordinate is the accumulator rather than the index, which gives a prediction: the benefit of the OOD-helping mechanisms should localise to the channels whose period exceeds the trained range of S_t, and ablating those channels should cost *less* at extrapolated length because they are already being read out of distribution.

The precondition holds — at the training length 8–17 of 64 channels never complete a cycle of their own period, and at 8× that length only 1–4 do — so the mechanism could operate. It does not. Ablating the 32 lowest-frequency channels is free at training length (−0.001) and costs 0.14–0.25 at $T = 1024$, on every arm whose baseline is off the floor. **Those channels are more load-bearing at extrapolated length, not less**, which is the opposite of the prediction. Damage is localised in frequency — high-frequency ablation costs 0.42–0.57 — but the plainer reading wins: the low-frequency channels carry position at map scale, and coarse position matters more when the walk covers more of the map.

13.7 F3: one quantity accounts for both

All four arms start from the same accumulator range at training length (88–94) and separate entirely by how fast it grows with sequence length, range$(S) \sim T^{\alpha}$:

Table 16: The accumulator's growth law. All four arms start from the same range at training length and separate by exponent.

arm	T=128	T=1024	α	opposition score
signed, r=4	91.8	269.7	**0.518**	0.125
signed, r=4 (rank sweep)	94.1	279.7	**0.524**	0.092
signed, r=2	88.5	320.5	0.619	0.495
monotone, r=4	91.5	649.8	**0.943**	1.849

A signed, well-conditioned increment gives $\alpha \approx 0.52$: the accumulator performs a **diffusive random walk**. A monotone one gives $\alpha \approx 0.94$: **ballistic drift**. The last column is the opposition score of §13.1; across the four arms $r(\text{opposition}, \alpha) = +0.9995$.

What α is worth, stated honestly. It is an *observable*, not a knob (see §17), and it is very nearly collinear with the opposition score: $r = +0.9995$ across these arms. Opposition is read straight off the learned action code, is simpler, and is easier to interpret, so α **carries almost no information that opposition does not**, and nothing in this review rests on α that could not have rested on opposition.

Its actual contribution is economy: it makes the sign result and the rank result *one* finding instead of two, at two severities. That is explanatory value, not predictive value — it does not tell anyone what to build. And its one candidate prospective use, predicting degradation at length, is precisely the link §17 records as unestablished.

One narrow use survives, and it is the reason to keep measuring it. Opposition needs *labelled opposite actions*; on text there are none, so it cannot be computed there at all. α needs only trajectories. So α is the form the cancellation measurement takes where the action code is not labelled — which is the setting the whole content-dependent-phase literature actually works in. That measurement has not been made: the language trainer here saves metrics and no model checkpoints, so it needs a retrain.

This is one mechanism for two separate results. How completely opposite actions cancel sets whether the accumulator diffuses or drifts; that sets how fast it leaves the range it was trained on; and that co-varies with the rate of degradation with length. The last link is the weak one and is stated as association rather than cause: θ enters through $\cos, \sin$, so a larger accumulator is not *prima facie* out of distribution at all, and the standard account of why it would be — untrained low-frequency channels — is refuted here (§13.6). A skewed rank-2 basis (opposition 0.495) and a monotone increment (1.849) are the same failure at different severities. The relation is correlational across four arms, and $\alpha \approx 0.5$ for a signed walk is unsurprising in itself; what it rests on is the spread to 0.94 and the agreement with an independently measured quantity.

13.8 F3, continued: the crossover

Every measurement above is on a task that wants a *map*. That makes "a signed increment is better" and "a signed increment suits this task" indistinguishable, and it is the obvious objection to §8. The test is a task whose answer depends on elapsed count rather than net displacement.

The task. A stream of symbols carries interleaved queries; a query names an offset k and the model must retrieve the k-th most recent *content* symbol, where uncounted filler tokens are interleaved at rate 0.5. It is a retrieval, not a decode — asking a MapFormer to *report* elapsed time reproduces a known failure, since the code makes positions comparable and not readable. The filler is what makes k a *contextual* position in CoPE's sense: the token distance back to the answer is

129.7 ± 10.3 at $k = 64$, so no fixed offset addresses it. Without the filler the task is index retrieval in disguise, and measured that way an index code scores 1.000 while signed and monotone tie. Six arms $\times$ eight seeds, one batch, $k_{\max} = 64$, trained at $T = 1024$; chance 0.0625, shortcut gates at chance, and doubling the budget moves the weakest arm by -0.009.

Table 17: The crossover. The cost of constraining the increment to be monotone, paired per seed, on a task that wants a map and one that wants a clock.

	torus (a map)	recency (a clock)	
monotone − signed, raw	−0.280	−0.004	
monotone − signed, loss-matched	−0.215	+0.026	both inside MDE
seeds	12/12 negative	4/8	

Giving up cancellation costs a fifth to a quarter of accuracy on the map task and **nothing measurable** on the clock task: an interaction of about +0.28. The accumulator says the same thing more sharply. The unconstrained arm may adopt either code, and adopts a different one per task, while every arm that *cannot* choose sits at the same exponent on both:

Table 18: The growth exponent is a readout of what the task demanded. Only the unconstrained arm can move, and it does.

arm	α (torus)	α (recency)	Δ
signed, unconstrained	$\mathbf{0.591 \pm 0.028}$	$\mathbf{0.967 \pm 0.009}$	**+0.376**
$\lvert\Delta\rvert$ (monotone)	1.010	0.976	−0.033
softplus (monotone)	1.005	1.005	+0.000
CARoPE form (monotone)	1.003	0.992	−0.011

$n = 12$ and 8; the shift has a standard error of 0.009. The constrained rows are the control that attributes it to the *choice* rather than to the two tasks' statistics.

Read this as corroboration, not as the finding. **The result is the accuracy interaction above** — forcing monotonicity costs 0.28 on one task and nothing on the other. The α table says the same thing measured on the learned code instead of on held-out accuracy, which is worth having because the two could have disagreed. It is not independent evidence, and given $r(\text{opposition}, \alpha) = 0.9995$ an opposition table would have said it equally well.

What supplies the clock, established by intervention. The arm does not make every increment positive — the negative fraction of Δ barely moves, $0.498 \to 0.478$. It learns a **content gate**: large increments on counted tokens, near-zero on filler, present in 8/8 seeds per head. Gate *strength* does not predict the outcome ($r = -0.34$ across a near-ceiling range), so this was settled by eval-only intervention rather than by more measuring. Removing the filler increments is a no-op ($1.000 \to 0.9997$); removing the counted ones destroys the task (0.068), which is the positive control. The decisive contrast holds magnitude fixed: replacing Δ with a *constant* on content alone scores 0.783, and with a constant on *every* token — same total drift, differing only in which tokens it lands on — 0.189, a gap of +0.594 at 8/8 seeds. A constant increment on content recovers most of the baseline, so the gate is most of the mechanism rather than a component of it.[1]

[1] An earlier condition that simply made the filler count also collapsed, apparently decisively. It is confounded and is not counted: a control that changed *only* the accumulator's scale, still counting content alone, collapsed just as hard. This model is acutely sensitive to θ's magnitude, so any intervention that moves it is destructive for reasons unrelated to counting. The magnitude-matched pair was added afterwards, as a control for that confound.

And a fixed index code cannot do it at all. The path-integrated arms reach 0.96–1.00 and the index arms 0.234, a gap of +0.750 at 8/8 seeds — larger than the navigation effect of §13.4. The per-offset curve says why: the index arms solve $k = 1$ (0.99) and collapse from $k = 8$ on (0.11–0.33), while every path-integrated arm is flat out to $k = 64$. This reproduces CoPE's own claim in a different architecture — against contextual position, a relative code's best is a decaying attention — and is cited here as an anchor rather than as a new result. Note also that a token clock built inside this architecture (0.189, above) lands *below* the index arms, which is what it is.

13.9 Where MapPoPE lands

This review began from a concrete question — MapFormer supplies a content-dependent *phase*, PoPE a content-dependent *magnitude*, and the coordinates are disjoint, so what does imposing both buy? — and the frame exists to answer it in a way that generalises. The answer is mixed, and the shape of the mixture is the useful part.

Table 19: MapPoPE's record. It is the strongest arm on the task the phase mechanism was designed for, and loses to a pure index code on the task where path integration itself fails.

task	MapPoPE	best alternative	reading
torus revisit, held-out map (n=8)	$\mathbf{0.994 \pm 0.017}$	MapEM-os 0.987; MapFormer 0.967	**best measured**
torus, OOD $l = 2048$ (n=8)	$\mathbf{0.970 \pm 0.028}$	MapFormer 0.854	**best measured**
PoPE's contribution, loss-matched, 4 grids (n=8)	+0.077 to +0.095	—	helps; *flat* in octaves, so the frequency-count account is refuted
MiniGrid DoorKey-16×16, T=1024 (n=8)	0.942 ± 0.017	**PoPE + *index*** 0.955 ± 0.003	**loses to an index code**
compositional transfer (n=8)	0.429 ± 0.117	MapFormer+hier 0.415 ± 0.096	tied
enwik8 bpc (n=3)	1.3786 (lowest)	PoPE+index 1.3806	numerically best, **not established**: the margin is under its own checkpoint sd

Three things follow, and the third is the one worth carrying away.

It is the best arm on the task the phase mechanism was built for, at the training length and at 16× it. That is a real result and it is where the combination earns its place.

It is not a general improvement. On MiniGrid it is beaten by PoPE with an *index* phase, because rotation-based actions defeat a fixed per-token increment (§13.4) — so the phase half is a liability exactly where path integration itself is. And on text the margin is inside its own noise. A combination inherits its components' failure modes, not only their strengths.

It fills a cell in the table without supplying what that cell promises. MapPoPE reads as "both slots" in Table 4, and §8 says a system needing map *and* clock must spend both. But PoPE's magnitude is *instantaneous* — read off the current token, not accumulated — so it adds no second accumulator, and measured, it does not move the growth exponent at all ($\alpha = 0.689$ against a baseline 0.710, inside its MDE). MapPoPE is two mechanisms in the phase-and-magnitude slots that supplies *one* answer. The genuinely unoccupied configuration is a signed accumulated phase beside an *accumulated* content-dependent magnitude in softmax attention, and §15 says what to build there.

14 Choosing a mechanism

What a task demands of a positional code, which mechanisms supply it, and which trade-offs are forced rather than incidental.

14.1 Three hard trade-offs

(T1) Commutativity buys the scan and costs path-dependence. With $K = 1$ and a linear increment the phase depends on the multiset of tokens traversed, the prefix product collapses to a prefix sum, and inference is $O(\log T)$ parallel. The same property makes displacement-under-accumulated-heading unrepresentable: a fixed per-token increment cannot express "forward" when forward depends on where you are facing. Measured both ways — the position effect falls from +0.438 to +0.050 under turn/turn/forward actions, and a mechanism that abandons the group law pays for it in the scan (§10). You may have one or the other.

(T2) A signed increment buys a map and gives up a clock. The accumulator's growth exponent is set by how completely opposite actions cancel, but the exponent is a symptom and the substance is §8: a signed increment makes the interval sum a *net displacement*, invariant to route and blind to elapsed time, and a monotone one makes it a *path length*, injective in time and blind to place. Neither dominates. On navigation the signed form wins outright (at matched training loss the monotone one does not beat a plain index clock anywhere); on language the monotone form is the correct object, and the signed one would be discarding the very distinction position exists to make.

(T3) Compressing the memory makes decay compulsory; rewriting it forfeits T1. Both halves are argued in §11: the primal form saturates so something must evict, while the dual form renormalises and may decline a decay term; and a memory that must *overwrite* needs a delta rule, whose transition matrix is a reflection rather than a scalar.

14.2 What a task demands, and what supplies it

14.3 What follows for choosing a model

On language, most of this is invisible, for the reason §8 gives: order is nearly all that is demanded, and a clock is the right object for it. The exception is worth naming, because it is the one place a text task makes a sharp demand: *contextual* counting — the k-th most recent item of some selected class — is a clock a fixed index cannot supply, and the gap there (+0.750, §13.8) is larger than anything measured on navigation. It is also the demand CoPE was built for.

On navigation and any task with a metric structure, the sign and the conditioning of the increment are the whole game. They are worth more per parameter than anything else measured here, and both dominate the choice of positional *encoding* (§13).

One combination is ruled out rather than merely absent. A mechanism with a bounded recurrent state, signed transport *and* rewritable memory does not exist, and T1 says the last two cannot both be had while keeping the parallel scan.

15 Predictions

The frame is worth something only if it says what to try next. These follow from the axes rather than from the measurements, and none has been run. Each is stated with what would refute it, so

Table 20: What a task demands of a positional code, and what supplies it.

if the task needs ...	it requires	and these supply it
order only — which token came first	any monotone clock	RoPE, ALiBi, CoPE, CARoPE, GRAPE-AP, every gated linear RNN
signed displacement — parity, net motion, state tracking	a signed increment	MapFormer, Selective RoPE, Mamba-3; DeltaNet/PaTH/RWKV-7 with $\beta > 1$. **Not** CARoPE, GRAPE-AP or CoPE
absolute location in a bounded map	signed **and** well-conditioned ($r \geq 4$)	MapFormer at $r \geq 4$; $r = 2$ degrades because its basis is skewed, not because it lacks capacity
holding position far past the training length	a diffusive accumulator, $\alpha \approx 0.5$	follows from cancellation (T2), though the route from α to degradation is association and not cause; nothing in Table 4 bounds the accumulator, and the two obvious ways to are a no-op or break additivity
path-dependent actions (turn/forward)	non-abelian transport, **or** an allocentric recoding of the input	PaTH, DeltaNet, RWKV-7, TEM-t — none evaluated on navigation. Recoding the action to absolute displacement restores +0.488 and keeps the scan
overwriting stale associations	a delta rule	DeltaNet, PaTH, RWKV-7; forfeits the parallel scan
contextual counting — the k-th most recent noun, sentence, or selected event	a **content-gated** monotone increment	CoPE, CARoPE, GRAPE-AP; and a signed MapFormer, which *learns* the gate rather than being given it. **Not** any fixed index code: RoPE reaches 0.234 where the gated arms reach 1.000
exact recall over a window	the dual form; keep the items	softmax attention. A lossy summary is not a sufficient statistic, which is why hierarchy loses here and wins on compositional transfer
bounded state, linear time	abelian transport	every row of Table 4; measured 2.6–3.3× over a 16× length increase

that the section is falsifiable rather than a wish list.

15.1 Mechanisms on tasks they have not been evaluated on

Table 21: Untested deployments the frame implies. None of these experiments exists.

mechanism	task	why the frame predicts it	refuted by
PaTH, DeltaNet, RWKV-7	navigation with **rotation** actions	non-abelian transport is exactly what turn/forward needs, and a fixed per-token increment cannot represent it: MapFormer falls from +0.438 to +0.050 there	landing at MapFormer's rotate level, which would mean the deficit is not about commutativity
MapFormer	**state tracking** (A_5, S_5), formal languages	a signed increment is a parity register — measured +0.316 at L=16 — and it keeps the parallel scan, which the delta-rule family forfeits	failing where DeltaNet succeeds, at matched budget
Mamba-3	**cognitive-map** navigation	the only mechanism holding both slots with a signed rotation; should be the strongest non-attention arm	falling to Mamba-2/GLA's level, which would mean the rotation is not what navigation uses
LieRE	navigation with rotation actions	its generators need not commute, so it is the one *positional* code that can express a heading-dependent displacement	needing a commuting special case to train at all
CoPE, CARoPE, GRAPE-AP	**navigation**	they are monotone, so §8 predicts they fail on a map task specifically, not generally	matching a signed phase on the torus
MapFormer / MapEM	transfer across a change of **structure** — new topology, new action space, a graph reused across environments	this is TEM's strongest claim and the one §10.1 does *not* test: everything measured here is a new instance of the same structure	the structural code failing to survive a topology change, which would confine the factorisation claim to re-instantiation
any structural code	**sample efficiency**, not final accuracy	if a where transfers, a new environment should need fewer episodes, not merely reach a higher ceiling. No learning curve has been measured here	equal sample efficiency with a higher asymptote, which would make the separation a capacity effect rather than a transfer one

15.2 What to borrow, specifically, for the what/where separation

If the goal is a cleaner separation of the structural code from the sensory one, the table says which mechanisms have a part worth taking and which do not.

Take CoPE's gate, but not its sign. MapFormer's separator is a linear bottleneck: Δ must reach zero on observations by cancellation in $W_{\text{out}}W_{\text{in}}$, and nothing makes zero a natural output. CoPE's is a sigmoid, for which zero is the resting state, and its entire purpose is deciding *which tokens count*. The evidence that this is the right borrow is that MapFormer re-derives it unprompted: on contextual counting an unconstrained increment learns a content gate in 8/8 seeds, and a magnitude-matched intervention shows the gate is most of the mechanism rather than a component of it (§13.8). But CoPE's gate is $\sigma(\cdot) \in (0, 1)$ and therefore non-negative, which by §8 builds a clock and not a map. The two mechanisms answer *different* questions — a gate says *whether* a token moves you, a sign says *which way* — and CoPE has only the first. The combination to build is a gate on a *signed* increment. One cost is worth stating: CoPE's gate is computed per query–key *pair*, which is why it has no θ_t at all and sits outside the frame; a *per-token* gate is the weaker but cheap version that keeps the prefix scan.

Read PoPE as the what-channel, which reframes MapPoPE. §13.9 treats PoPE's magnitude being *instantaneous* as a deficiency, because it supplies no second accumulator. Under the factorisation question it is the opposite: an instantaneous content-driven magnitude carries sensory information *inside* the positional operator while being structurally unable to corrupt the structural code, and that is measured — adding it does not move the growth exponent (0.689 against 0.710, inside its MDE). So MapPoPE is already a what/where split within the operator: an accumulated signed phase for the where, a non-accumulated content magnitude for the what. It is also the best arm measured on the torus task (0.994 against 0.967). The same property is a defect for a system needing a clock and a virtue for one needing a clean separation, which is the sort of thing only the frame makes visible.

Do not borrow Selective RoPE's gate for this. It is the obvious candidate — a sigmoid on the phase increment — and it was tested directly. It does not suppress by token class: on the torus it gives 0.560 on actions against 0.415 on observations, a ratio of 1.35× and nowhere near suppression, and on parity it is 1.54–1.70× in a setting where the gate *hurts*. Whatever it does, it is not separating what from where.

Do not borrow the forget gate for this either. It sits in the magnitude slot and is monotone, so it is a clock (§8); it suppresses what is *remembered*, not what updates position. It is the right borrow for a different problem, listed below.

Algebraic PE is the one to take when the structure is known — and it is the most relevant row in the table to a *relational* goal. It maps "the algebraic specification of a domain" to orthogonal operators so that the model "upholds its desired structural properties", and it covers sequences, grids, **trees, and their compositions**. Two consequences. It expresses structures MapFormer's abelian $SO(2)$ cannot — a tree is not a product of circles — so it is the natural vehicle for the one prediction in Table 21 that the factorisation thesis most needs and this work does not test, transfer across a change of *structure* rather than of instance. And it is the honest oracle for the separation itself: where MapFormer must *infer* the where from the action stream, Algebraic PE is handed it, so the gap between them prices what inference costs.

TAPE's idea is already tested here, and it is a null. TAPE contextualises the positional encoding "with sequence content across layers", refining it at each update — i.e. content-driven *re-estimation* of the where along depth. That is the same object as the refine-θ variant measured in this project, which carried and corrected the position estimate at each pass. It does nothing: $-0.001/-0.011/+0.005$ at training length across three noise levels, with no slope in noise, and the learned gate settles near zero, so the optimiser *declines* to refine rather than being unable to. That is a mechanism answer and it applies to this borrow before it is attempted.

HGRN offers a third kind of separation: by timescale, bound to depth. Its forget rate carries a lower bound that "increases monotonically when moving up layers", so lower layers model short-range and upper layers long-range structure. That is neither a slot nor a gate but a third axis on which what and where could be divided, and it has a concrete form here: MapFormer fixes the same geometric ω ladder in every layer, and binding its range to depth instead is a free change with a published precedent. Untested.

Titans gates the other side. Its memory is written by *surprise* — selection on what is worth storing, not on what updates position. A gate on the where and a gate on the what are complementary, and only the first is discussed above.

And keep the bottleneck. Since W_{in} *is* the identification mechanism, its rank is the channel the whole distinction passes through, which is why $r = 2$'s skewed basis is a fact about the separation and $r = 4$ repairs it for 384 parameters (§13.2).

15.3 Combinations worth building

Table 22: Combinations the frame identifies, ordered by how much evidence already points at them. None has been built.

combination	what it fixes	the evidence pointing at it	refuted by
signed accumulated phase + accumulated content-dependent magnitude (MapFormer + a forget gate), in softmax attention	the empty cell: a map *and* a clock at once. Mamba-3 holds it in an SSM; nothing holds it in softmax attention	the forget gate's own accumulator is ballistic ($\alpha = +0.956$) beside the phase's 0.6, so the clock is already there and unexploited; and it explains why the gate needs a *live* λ but not decay	no gain on a task needing both, i.e. recency-dependent navigation, which does not yet exist
MapPoPE + a forget gate	MapPoPE's actual deficit — it reads as both slots and supplies one answer (§13.9)	α unchanged by PoPE (0.689 vs 0.710), so the second accumulator is genuinely absent	α still unmoved, which would mean the magnitude slot cannot host a clock at all
explicit content gate + signed phase — as an oracle **control**, not a proposal	measures what learning the what/where split costs, against a design whose stated purpose is to learn it	on contextual counting an *unconstrained* signed arm learns the gate by itself, and a constant increment on the gated class recovers 0.783 of 1.000 (§13.8)	the explicit gate not beating the learned one, which would mean the learning is free
signed phase + non-abelian transport	rotation action spaces, without the allocentric recoding that needs known heading	recoding restores +0.488, so the information is sufficient; the question is whether the mechanism can carry it	the parallel-scan loss costing more than the accuracy gained
rank r=4 + weight sharing across depth	reliability rather than peak accuracy	measured together: 0.986 ± 0.020 with 8/8 seeds above 0.941, against 0.416 for either alone — the one super-additive pair found here	— (this one is measured; it is listed because it is the template the others are proposed on)

The ordering principle. Every combination above pairs mechanisms that occupy *disjoint* coordinates of (5). That is the frame's one practical prescription: mechanisms in the same slot compete and mechanisms in different slots may compose, and which of the two is happening can be read off the table before anything is trained. MapPoPE is the cautionary case — disjoint coordinates are necessary and not sufficient, because both of its halves answer the same question.

16 Conclusion

Three assumptions — linearity, translation invariance, continuity — reduce the space of positional mechanisms to the classification of a single generator, and the generator has exactly two live parts. Everything in the arena is a choice about where content enters those two parts, and everything that appears to escape does so by giving up one of the three assumptions, at a price that can be named: the classical relative methods give up continuity and with it any extrapolation; the delta-rule family gives up commutativity and with it the parallel scan; CoPE and DAPE give up per-token factorisation and with it any θ_t at all; and MesaNet, Titans and TAPE give up the accumulation structure itself.

The two parts are not two knobs. A signed accumulated phase is a map and a monotone one is a clock, and a system that needs both must spend both slots. That is why the monotone increments of CARoPE, GRAPE-AP and CoPE are correct on language and fail on navigation, and why the sign of the increment — which costs nothing, and which no paper in the positional-encoding literature varies deliberately — is worth more than any other ingredient measured here. The dichotomy is not only an interpretation: on a task whose answer depends on elapsed count rather than net displacement the cost of forcing a monotone increment falls from 0.28 to nothing measurable, and an architecture free to choose picks a different accumulator for each task (§13.8). The corollary for practice is that a mechanism does not have a quality, it has a match: what to spend on is decided by the task.

What the frame does not yet explain is the axis on which several mechanisms independently help: extrapolated length. The accumulator's growth law accounts for two of them and demonstrably not for the others — and no mechanism in the table bounds the accumulator at all, including the one that was assumed to. Nor is the growth law itself yet a cause: θ is periodic, so a larger accumulator is not obviously out of distribution, and the one published account of why it would be is refuted here. α unifies two results and reads off what a task demanded; it does not yet explain either. That is the gap worth building into.

17 Open questions

- **The input-side rank.** $r = 2$ is the worst setting and $r = 4$ fixes it for 384 parameters, but nothing explains why the optimiser finds a skewed basis at $r = 2$ and a well-conditioned one at $r = 4$ when the learned solution occupies a 2-plane either way.
- **The OOD-length axis, still the largest gap.** The accumulator's growth exponent accounts for the sign and rank results together (§13.7), and it predicts that bounding the accumulator should help at length. The obvious candidate for something that bounds it — the wrapped innovation of an InEKF-style correction — does not: it wraps the innovation and not the state, and the accumulator's range is unchanged (285.6 against 283.9). Measured, neither the forget gate nor PoPE changes α either (both inside their MDEs), so at least one other cause is at work on an axis where four separate mechanisms help. §8 proposes an account for the

forget gate — it restores the clock a signed phase gives up — of which one prediction is now cheap to run, since §13.8 supplies a task with explicit recency structure.

The obvious test of the growth law is malformed, and the real gap is prior to it. An earlier version of this section proposed varying α deliberately and checking that degradation follows. That cannot be done as stated. α is not a parameter of any model here — it is a statistic fitted to a trained one — and the two ways to move it both fail: interpolating from a signed to a monotone increment moves α but destroys the map at the same time, so on a map task it isolates nothing beyond the sign ablation; and bounding the accumulator is either a no-op, since θ enters through $\cos, \sin$ and θ and $\theta + 2\pi$ are already identical, or it breaks additivity and with it the interval-relativity that the whole frame rests on. Within this frame α appears not to be independently controllable: it is set by the cancellation, and the only lever on the cancellation is A5 and A6. That would make α a *re-description* of those two axes rather than a separate cause — which is consistent with its covering exactly those two mechanisms and none of the other four.

The prior question is why a growing accumulator should cost anything at all, given that the code is periodic. The natural answer — that low-frequency channels reach phases never trained — is the long-context literature's critical-dimension account, and it was imported and **refuted** (§13.6): ablating those channels costs *more* at extrapolated length, not less. So α's association with OOD degradation is established and its route is not. Until that is answered, α *unifies* the sign and rank results and *diagnoses* what a task demanded, and it does not explain.

- **Defective generators.** Jordan-RoPE occupies the cell; what polynomial factors in the lag are *for* remains open, and no navigation evaluation exists.
- **NoPE.** A causal decoder learns an implicit position without any operator at all, and in at least one controlled ablation it has the best perplexity of every encoding tested (Table 6). The frame classifies operators and has nothing to say about the case where the right operator is the identity; when an encoding helps on text, how much of the gain is over NoPE rather than over RoPE is usually not reported.
- **Sign on language.** A monotone clock cannot represent a -1 action, and on navigation that costs everything. On text HGRN's own ablation finds a data-dependent phase worse than a constant one. Whether the sign matters there, and for what, is untested — and §13.8 sharpens rather than settles it: on a symbolic clock task the sign is free to be given up at no cost, which is consistent with text needing a clock, but the arms there are trained from scratch on one synthetic stream. The published dataset that would settle it is Flip-Flop LM, which is that task at $k = 1$.
- **Non-commuting action spaces.** Rotation-based actions defeat a fixed per-token increment. Allocentric recoding sidesteps it wherever heading is known; the non-abelian mechanisms (PaTH, DeltaNet, RWKV-7) address it directly and have never been evaluated on navigation.

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